

CN4119 Final Year Design Project

Engineering Economic Analysis

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Recommended Readings

Turton et al.'s book (Reference 5) Chapter 9

Overview: Engineering Economic Analysis

- ❑ Investment, interest and time value of money

- ❑ Cash flow diagrams
 - Discrete
 - Cumulative

- ❑ Depreciation of capital investment

- ❑ Taxation and profits

- ❑ Typical CFD for a new project

Introduction

- ❑ Okay, capital and operating costs estimated - *what next?*
- ❑ An economic evaluation to determine whether:
 - Chosen process generates money
 - Chosen process is attractive compared with other alternative processes
- ❑ Several concepts and relations for profitability analysis
 - Investment, interest and time value of money
 - Depreciation and working capital
 - Revenue, tax, profit and cash flow

Investment & Time Value of Money

Investment and Interest

- ❑ Investment and interest
 - Investment to earn interest (or return)
 - By individuals and companies

Money, when invested, makes money

- ❑ Invest P \$ today for n years
- ❑ Interest/return of i \$/(\$·year) (say, 0.1 or 10% per year)

❑ Value/amount after n years:

➤ $F_n = P (1 + n i)$ assuming simple interest

➤ $F_n = P(1 + i)^n$ assuming annual compound Interest

Period	Capital at start of period	Interest paid during period	F_n at end of period
1	P	$P i$	$P + P i = P(1 + i)$
2	$P(1 + i)$	$P(1 + i)i$	$P(1 + i)^2$
3	$P(1 + i)^2$	$P(1 + i)^2 i$	$P(1 + i)^3$
.			
n	$P(1 + i)^{n-1}$	$P(1 + i)^{n-1} i$	$P(1 + i)^n$

Which interest – simple or annual compound, is better for the investor?

Time Value of Money

- ❑ Is \$10,000 at the **present time** better than the same amount at the **future time**?

Amount at the present time	Amount at the future time (say, 10 years from now)
\$10,000	$\$10,000 (1+0.1)^{10} = \$25,937$
$\$3,855 [= \$10,000 / (1+0.1)^{10}]$	\$10,000

Money today is worth more than that in the future

- ❑ Amount in year n (F_n) can be **discounted** (brought) to the present time

- ❑ Rearranging $F_n = P(1 + i)^n$ to:

$$P = \frac{F_n}{(1+i)^n}$$

- ❑ Time value of money refers to different values of the investment at different times due to the earning capability of money

- ❑ **Money can grow only if invested**

Interest rate → Discount rate

Example 9.3: Time Value of Money

You estimate that you need \$1150 to buy your dream bag in two years' time 😊 You have two choices: (1) Wait for two years before you take any action. (2) Invest \$1000 now which will provide \$1150 in two years. What would you do? And, what is the interest/discount rate?

Cash Flow Diagrams

Cash Flow Diagrams (CFDs)

- ❑ Show investment and profit (outward and inward cash flows) in each year of the project
- ❑ Discrete and cumulative CFDs
 - ***Discrete CFD***
 - Clear, unambiguous pictorial record of the value, type and timing of each transaction occurring during the life of a project
 - ***Cumulative CFD***
 - Keeps a running total of the cash flows occurring in a project

Example 9.10: Discrete CFD

You borrow \$1000, \$1200 and \$1500 from a bank at the end of years 1, 2 and 3 respectively. At the end of year 5, you make a payment of \$2000, and at the end of year 7 you pay off the loan in full.

Discrete CFD from your perspective

Discrete CFD from bank's perspective

Now, how will a plant's CFD look like?

Capital Depreciation

Depreciation of Capital Investment

- ❑ Accounts for decrease in value of equipment/plant with time

- Purchase and installation of equipment

=> **Negative cash flow** before plant is operational

- Equipment salvaged after plant is closed

=> **Positive cash flow** at end of plant life

- Difference between these costs

=> **Capital depreciation**

- ❑ A complex and often confusing subject!

- In general, an annual allowance with calculation government-controlled

- Larger depreciation in a given year => Smaller income tax => Greater profit

- ❑ Strategy to recover the investment in the plant and to decrease tax payable on profit

- A fraction of the capital depreciation is charged as an operating expense each year until the total capital depreciation has been charged

- ❑ **Depends on local tax laws**

- Amount and rate at which equipment may be depreciated set by government

Example 9.20: Capital Depreciation

Consider investing \$100,000 on a new machine that will yield additional before-tax profits of \$16,800 per year. The machine has a scrap value of \$2000 at the end of its useful operating life of 4 years.

- ❑ Capital depreciation = $\$100,000 - \$2000 = \$98,000$ over 4 years
- ❑ \$98,000 investment a long-term expense
- ❑ Owner should be able to deduct it as a legitimate operating expense
- ❑ Owner could have invested \$100,000 in a mutual fund that yields 10% per year before tax

- ❑ ***Depreciation is the method that government allows for businesses to obtain operating expense credits for their capital investments***

□ Let's tabulate the cash flows for both scenarios

Year	Invest in machine	Invest in mutual fund
0		
1		
2		
3		
4		
Total		

Working Capital

- ❑ Total capital investment = Fixed capital + Working capital
 - ❑ Fixed capital: FCI (total module cost or grass roots costs)
 - ❑ **Working capital (WC):** 'Float' of money required for starting up and financing the first few months of operation; this 'float' of money needs to be kept every year to maintain the inventories in advance; this 'float' is not needed at the end of the project
 - General rule of thumb: 15-20% of FCI
 - Can be fully recovered at the end of the project
- => Cannot be depreciated**

Types of Depreciation

- ❑ Depreciation calculations require the following:
 - Fixed capital investment ***excluding land*** (FCI_L)
 - Represents depreciable capital investment
 - Life of the equipment/plant (n years)
 - Specified by government authority
 - Time allowed for equipment depreciation
 - Salvage value, S
 - Fixed capital investment excluding land at the end of the plant life
 - A small fraction of the initial FCI

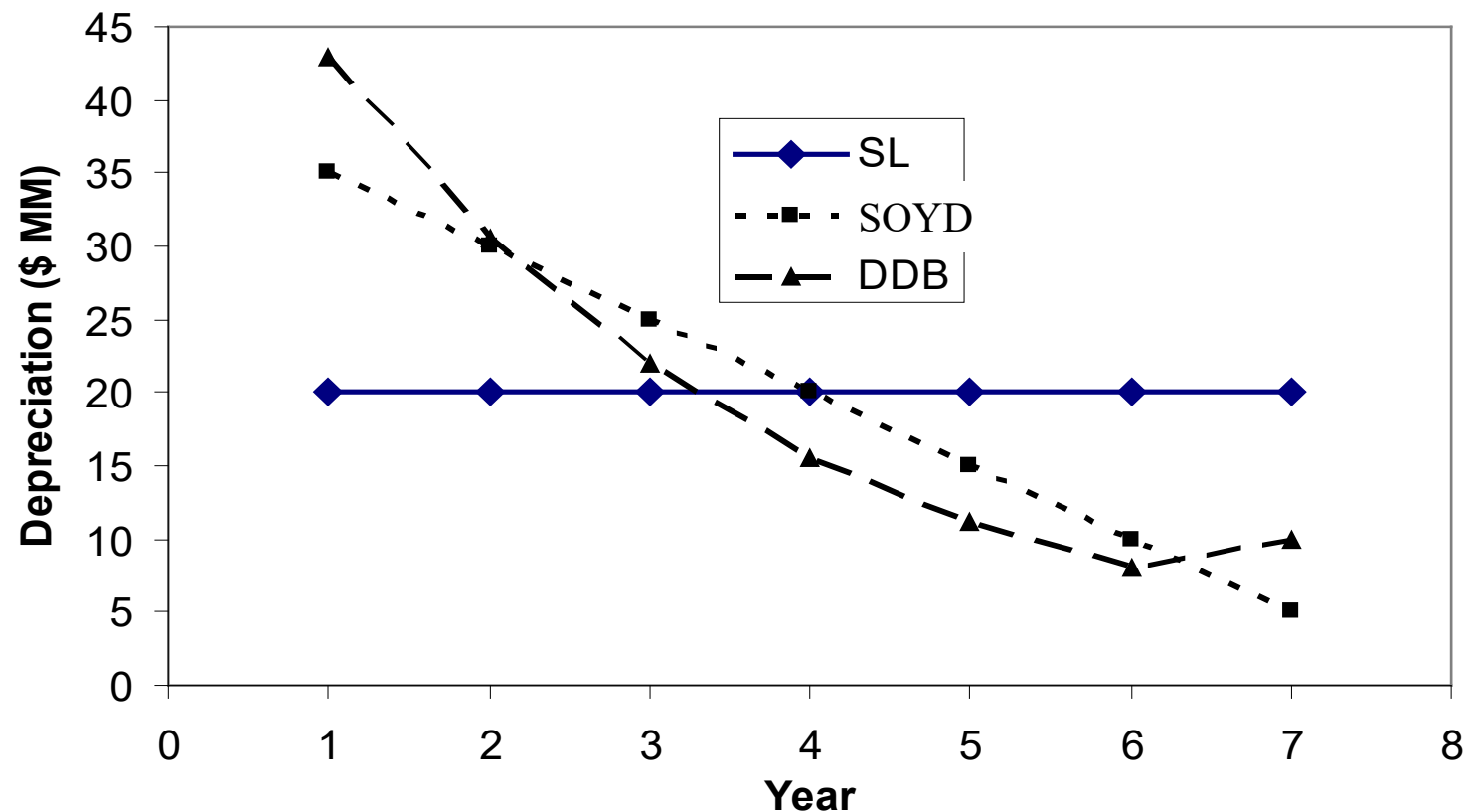
=> often assumed to be zero
 - Total allowed capital for depreciation,
 $D = FCI_L - S$
 - **Yearly depreciation**
 - Amount allowed in k th year is d_k

1. ***Straight line*** (SL)
depreciation
 - $d_k = D/\text{depreciation period}$
 - An equal amount of depreciation charged each year over the period allowed
2. Sum of the years digits (SOYD) depreciation
3. Double declining balance (DDB) depreciation

Example 9.21: Types of Depreciation

Fixed capital investment (excluding land cost) in a new project is \$150 million and the salvage value of the plant is \$10 million at the end of 7 years of equipment life. What is the depreciation according to the different methods?

Which method is better?



Profit and Cash Flow

Taxation, Profit and Cash Flow

- ❑ Revenue = R
- ❑ Expenses = $COM = COM_d + d \rightarrow$ Here, d is an operating expense allowance, not an actual expense incurred in that year

Where,

COM_d = Cost of manufacture excluding depreciation

d = Depreciation

- ❑ Tax rate = t
- ❑ Income tax = $(R - COM_d - d) t \rightarrow$ Company pays higher tax if d is 0 (that is, if not allowed legally)
- ❑ After-tax (net) profit = (Revenue – Expenses) – Tax = $(R - COM_d - d) - (R - COM_d - d)t$
- ❑ After-tax cash flow = $(R - COM_d - d) - (R - COM_d - d)t + d \rightarrow$ If d is not added back, cash flow will be lesser which is not correct

The two d 's cancel off $\Rightarrow$ no effect on actual cash flow

Simple Example

- ❑ Let's assume $R = 100$, $COM_d = 10$, $d = 15$, $t = 0.15$ (i.e., 15%)
- ❑ Before-tax profit = $(R - COM_d - d) = 100 - 10 - 15 = 75$
- ❑ Income tax = $(R - COM_d - d) t = 75 \times 0.15 = 11.25$
- ❑ After-tax (net) profit = $(R - COM_d - d) - (R - COM_d - d)t = 75 - 11.25 = 63.75$
- ❑ After-tax cash flow = $(R - COM_d - d) - (R - COM_d - d)t + d = 63.75 + 15 = 78.75$ ----
(1)
- ❑ Actual cash flow for company = $R - COM_d - \text{Income tax} = 100 - 10 - 11.25 = 78.75$ -----(2)

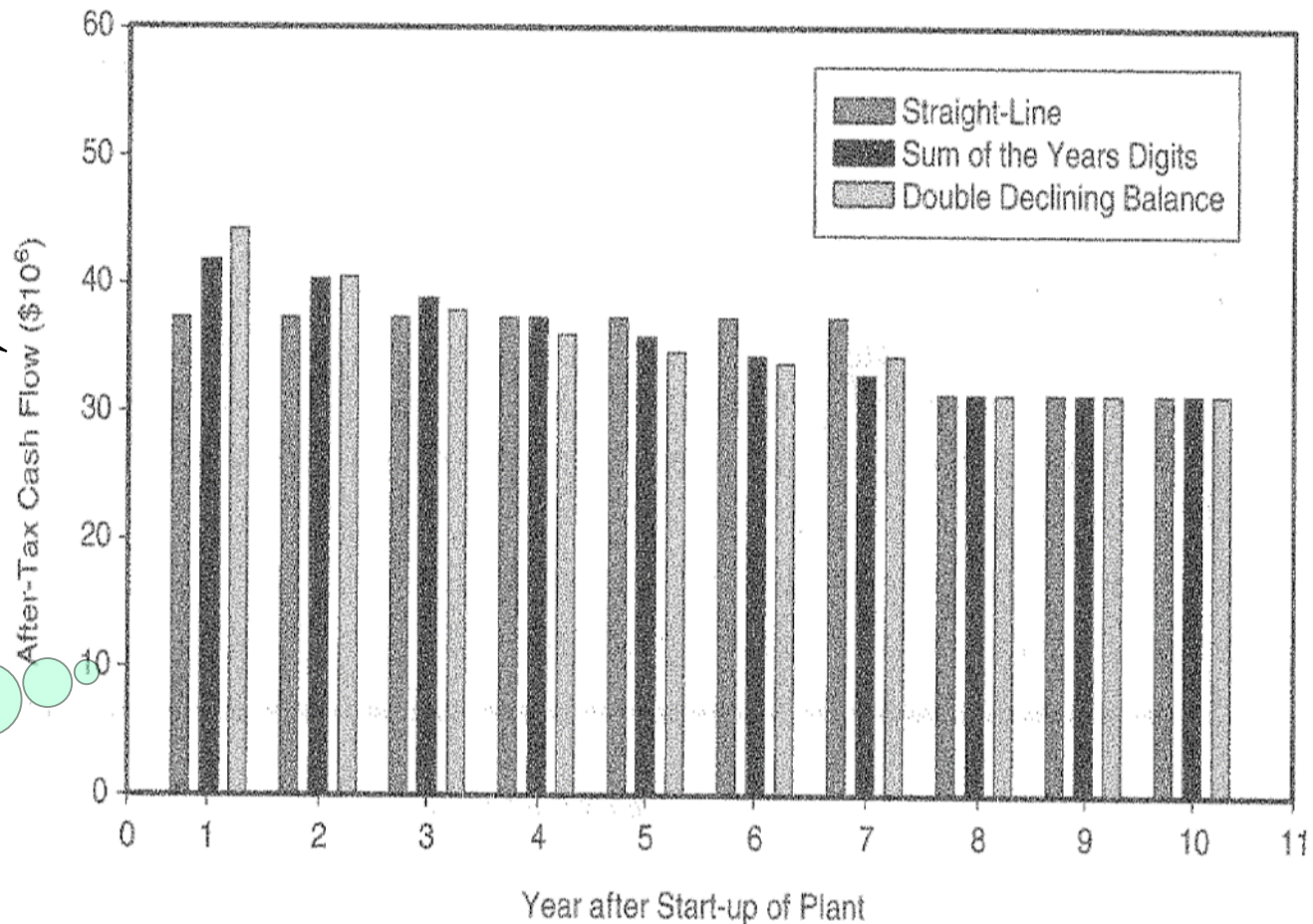
- We can see that (1) = (2) which is correct.
- If d was not added back in (1), after-tax cash flow would have been 63.75 which is not correct!

Example 9.23: Profit and Cash Flow

Calculation of profit and cash flow for a 10-year period after start-up of the plant.

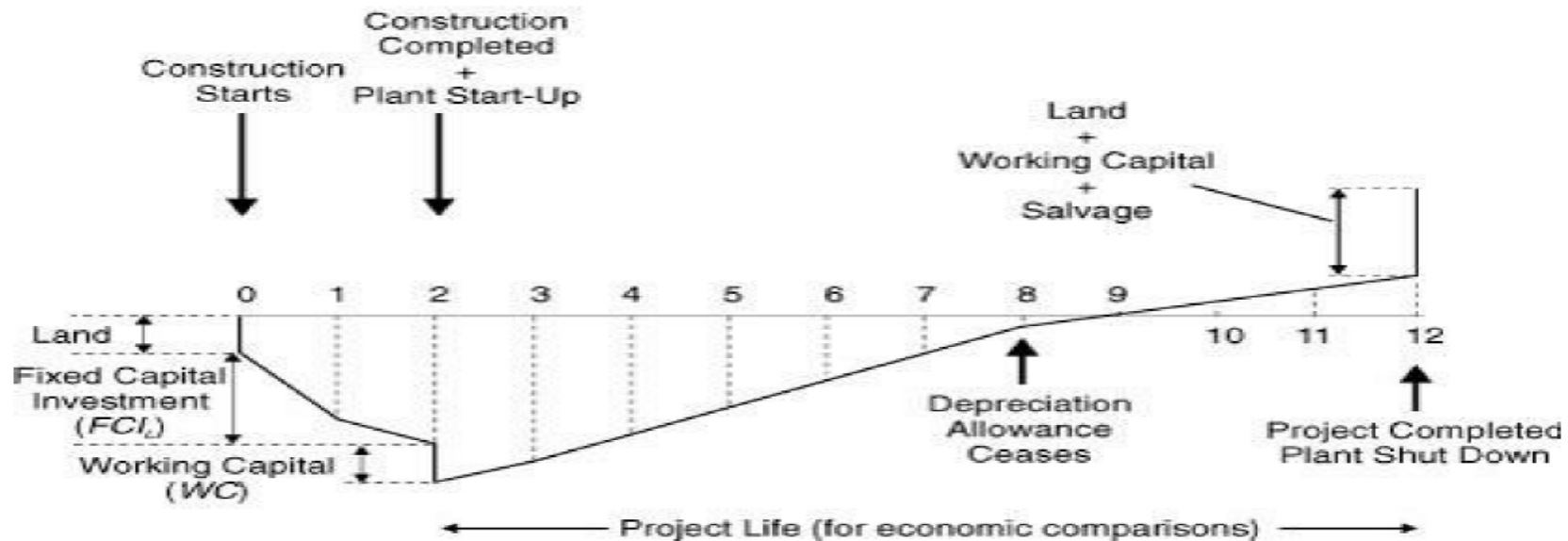
After-tax cash
flow

***Which
method is
attractive?***



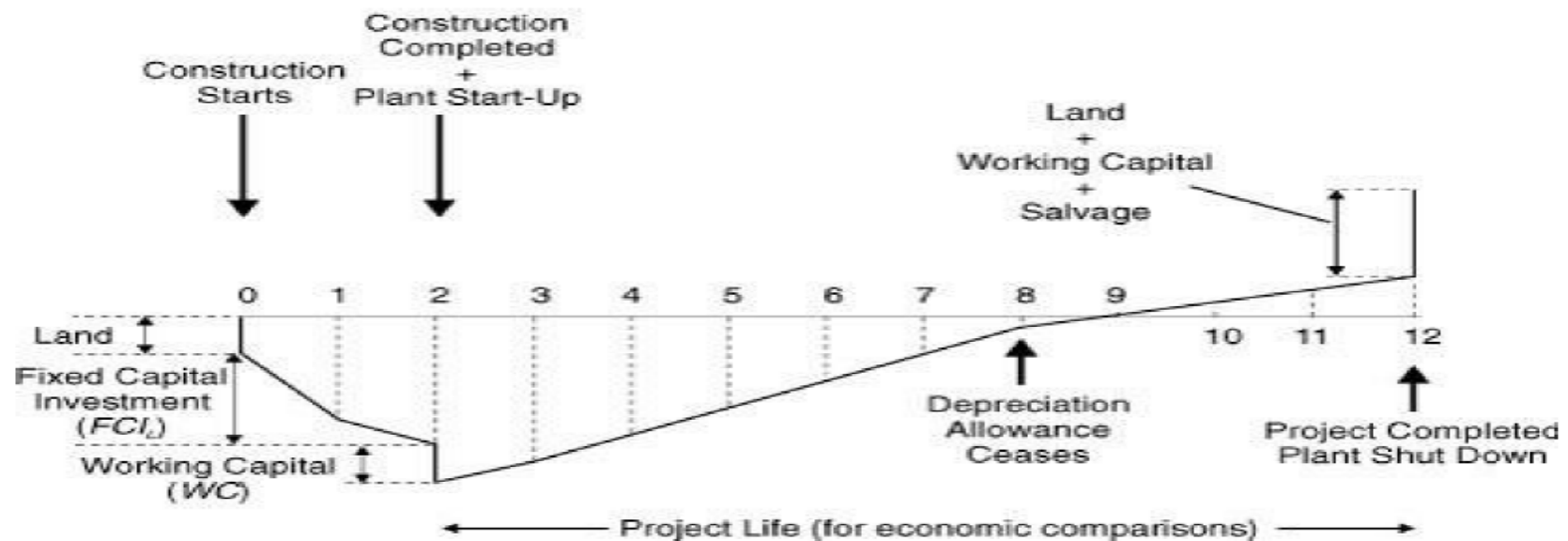
Typical Cumulative CFD

Typical Cumulative CFD for a New Project



- Land purchase made at time zero (at project start)
- Assume two-year construction phase
 - FCI
- Plant start-up at end of second year
 - WC

- After start-up, sales begin
 - Yearly cash flows become positive
 - Positive slope of cumulative CFD
 - Cash flows for earlier years of operation larger due to the depreciation allowance effect
 - Six years depreciation for the above figure



- 10 years of operation (2 to 12)
 - This is the project life n
 - Steadily rising cumulative cash flow
- At the end of 10 years of operation
 - Assume plant shut down
 - Land sold, WC recovered, salvage value for equipment obtained

=> vertical line in year 12

- ❑ Cumulative CFD is positive at the end
 - ❑ How do we ensure that the project is profitable?
- Is income earned during plant operation greater or lesser than initial investment?
- => Time value of money must be considered

Engineering Economic Analysis: Summary

- ❑ Money + Time = More money
 - Require resources to make an investment and time to allow the investment to grow
 - Revenue (i.e., additional money) generated when low-value materials are converted to high-value materials
- ❑ Cash flow diagrams to visualize timing of cash flows and to manage cash flows
- ❑ Depreciation, taxation and profit
- ❑ Typical cumulative CFD for a new project
 - Initial investment: Land, FCI, WC
 - Cumulative CFD must be positive in the end